A Comparison of the Effects of Acute Whole-Body Vibration for Relieving Vocal Fatigue in Young and Older Adults

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Summary: Purpose. Whole-body vibration (WBV) is a novel treatment to alleviate vocal fatigue, but the evidence in older adults is limited. This study applies WBV in young and older adults to explore how age and vibration duration may influence its relaxation effect.

Methods. Eighteen older females (mean age 65.56 years) and 18 young females (mean age 22.39 years) with healthy voice participated in karaoke singing to induce vocal fatigue, followed by 15 minutes of WBV. Participants' vocal changes were measured using i) the highest fundamental frequency produced, ii) self-perceived vocal effort, and iii) vocal fatigue level at six timepoints (prevocal loading, postvocal loading, post 5-minute WBV, post 10-minute WBV, post 15-minute WBV, and 15 minutes postWBV completion).

Results. A significant main effect of vibration duration ($P < 0.001$) and an interaction effect (age x vibration duration) ($P < 0.05$) were found for self-perceived vocal effort and vocal fatigue level. Both groups showed significant fatigue reduction after receiving 5 minutes of WBV, but recovery slowed down for older adults after receiving WBV for 10 minutes. A significant main effect of age ($P < 0.001$) was found for the highest fundamental frequency.

Conclusion. The results support that WBV may be used in both older and younger adults for vocal fatigue relief.

Key Words: Whole-body vibration—Vocal fatigue—Age effects—Vibrational dosage.

INTRODUCTION

As a person ages, the laryngeal system experiences various anatomical and physiological changes. There can be atrophy of the intrinsic laryngeal muscles, loss of intrinsic laryngeal muscle elasticity, calcification and ossification of laryngeal cartilages, and mucous gland degeneration, leading to vocal fold bowing and glottic insufficiency.[1–3] Laryngeal agility reduces and vocal fold vibrations become more irregular and unstable.[4] These age-related structural and functional degenerations in the larynx are called presbylarynx, or aging larynx.

Deterioration in voice qualities and voice-related quality of life associated with presbylarynx is referred as presbyphonia.[5,6] Presbyphonia is a diagnosis of exclusion.[6–9] The diagnosis of presbyphonia is made when medical conditions and vocal pathologic abnormalities other than aging larynx are excluded.[8] Nevertheless, one should note that although presbyphonia is attributed to aging, it is not always predicted by obvious age-related laryngeal change.[5,7,10] Presbyphonia is common among the older populations.[6,11,12] Wang et al[13] conducted a systematic review with meta-analysis to examine the prevalence of voice disorders in older adults. The prevalence of presbyphonia for those who are independent living residents in the community was 15.2%. Higher prevalence of 33.03% was noted in older adults who were institutionalized.

The presence of presbylarynx, together with the age-related declines in the respiratory system, including the reduction in chest wall compliance, lung elasticity and recoil forces, as well as weakened inspiratory and expiratory muscles, results in the development of compensatory behaviors during speech production.[14,15] Respiratory effort as well as supraglottic hyperfunction to compensate for the glottic insufficiency are common in this population. In order to generate sufficient subglottal pressure for speech, older adults tend to initiate speech at higher lung volume to compensate for the reduced recoil pressure, especially when speaking long utterances or when speaking loudly against background noise.[16] Such speech initiation requires greater effort from the inspiratory muscles, which can lead to fatigue.[16,17] Vocal fatigue is one major vocal complaint frequently reported by older adults and is typically observed after prolonged voicing.[18] Vocal fatigue brings negative impacts to the older adult's quality of life, reducing their social interaction and increasing their dependency.[19]

Whole-body vibration (WBV) involves transmission of vertical/horizontal oscillatory motion to the human body from a vibrating foot platform.[20] Yiu et al[21–23] attempted to apply WBV onto relieving vocal fatigue. They proposed two hypotheses to explain how vocal fatigue relief could be achieved with WBV. The first hypothesis is drawn based on the conceptualization that the larynx is supported by connective tissues of ligaments, tendons, and cartilaginous joints. These connective tissues serve as springs holding the larynx. The WBV transmits the vertical vibration from the vibrating foot platform to the larynx, bringing the larynx bounces vertically up and down. Such vertical vibrations lead to periodically stretching of the larynx (leading to

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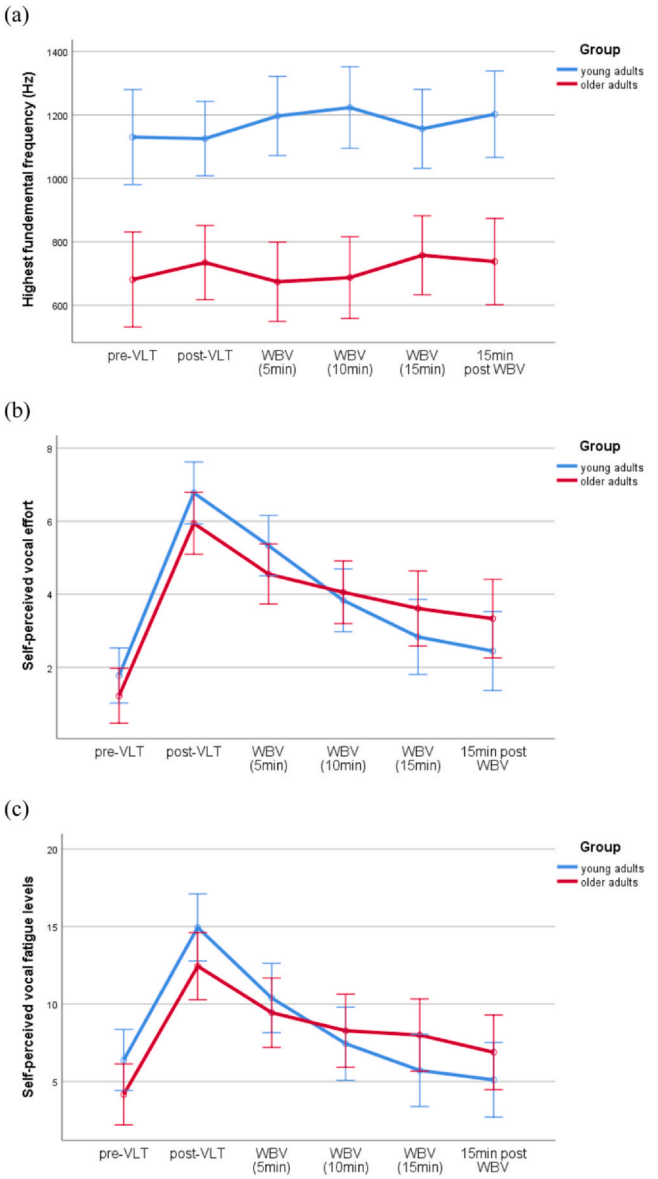


FIGURE 1. Mean change of **(a)** the highest fundamental frequency, **(b)** self-perceived vocal effort, and **(c)** self-perceived vocal fatigue levels across the six timepoints.

vocal fold lateralization) and compression of the larynx (leading to vocal fold medialization). This periodic vocal fold lateralization and medialization exhibits a massaging effect to the vocal folds and relieves excessive laryngeal muscle tension. The second hypothesis is that WBV enhances blood circulation around the larynx so that metabolic waste can be disposed of from the fatigued laryngeal muscles quicker, relieving vocal fatigue.

There is emerging evidence on the use of WBV for relieving vocal fatigue. Yeung[24] examined the effects of WBV on maximum phonation time in 30 young adults after a prolonged singing task. She found a significant increase ($P < 0.005$) in maximum phonation time in the WBV group, which had undergone 10 minutes of WBV (30%, 5 Hz), as compared with the vocal rest group. Her findings align with the study by Yiu et al,[22] which compared the effects of WBV, localized vibration, and voice rest for relieving vocal fatigue in 44 young vocally healthy adults after a prolonged singing task. The effect of a 10-minute WBV was comparable to localized vibration but superior to voice rest. Significant increase in the highest fundamental frequency produced and decrease in self-perceived vocal fatigue were noted in the WBV group ($P < 0.002$). Similar positive effects of WBV for relieving vocal fatigue were reported in a subsequent study by Yiu and Lee.[23] That study compared 10 minutes of machine-generated WBV to self-induced vibration for relieving vocal fatigue in 24 female young adults. Significant increase in the highest fundamental frequency was only noted with machine-generated WBV ($P < 0.02$).

Despite the encouraging findings, existing studies using WBV to relieve vocal fatigue have been restricted to young adults only. Using WBV to relieve vocal fatigue in older adults has not been investigated. Nevertheless, WBV has been extensively studied by physiotherapists for its use in improving muscle strength and functions as well as how the effect manifests differently in young and older adults, though the results have been inconclusive.[25–28] For instance, Lienhard et al[27] reported that WBV induced more lower limb muscle activity in older adults than young adults, whereas Delafontaine et al[25] found WBV to be only effective in young but not older adults in enhancing dynamic stability. Based on the physiotherapy literature, it would be interesting to compare how young and older adults with vocal fatigue respond differently to WBV. Due to their reduced laryngeal elasticity and blood flow rate, older adults may experience slower recovery from vocal fatigue.[29] On the other hand, it could also be argued that since older adults have poorer vocal performance and are more prone to fatigue, their potential for vocal recovery is higher.[30] As such, how age may influence the relaxation effect of WBV is yet to be studied.

Another variable that has been under-researched is dosage, which is referred to as vibration duration in the current study. The three reviewed WBV studies[22–24] included only one bout of 10-minute vibration. Though significant vocal fatigue reduction was shown, it is unknown if such duration is optimal for both young and older adults. Specifically, given the biological changes in an aging larynx, the vibration duration needed to show significant improvement in older adults may be different from young adults. Physiotherapy research has reported a wide range of vibration duration for older adults, from 30 seconds to 10 minutes.[31] There is inconclusive evidence on whether longer or shorter vibration duration is favored for inducing immediate neuromuscular changes.[32] Bosco et al[33] found that 10-minute WBV (26 Hz, 10 mm) improved velocity, force, and power of leg muscles. Tseng et al[28] used 5-minute WBV and reported an increase of muscle activity in young and older adults. Torvinen et al,[34] however, found no improvement in muscular function in healthy young adults after 4 minutes of WBV (2 mm). These findings

suggested that more research needs to be done to evaluate how vibration duration may impact the effect of WBV, particularly on vocal fatigue relief.

WBV can be particularly relevant to the older population for the following reasons: first, WBV is an external, passive, and noninvasive exercise. Compared with other voice training modalities such as semi-occluded vocal tract exercises and localized vibration to relieve vocal fatigue, WBV requires less degree of cognitive and coordinative skills of the user. For example, when using a localized vibrator, the user needs to apply the vibrator at designated placements in the perilaryngeal area (see[22,35–37] for protocols of localized vibrator placements). Vocal exercises such as straw phonation in water exercise requires the individual's kinesthetic awareness to sense the water bubbling during practice.[38,39] These exercises may pose challenges for older adults with declining fine motor functioning and cognitive capacity. Other vocal exercises such as vocal function exercises and phonation resistance training exercises have shown to be useful for mitigating aging voice.[3,40,41] However, hearing impairments and voice problems are common in the older population and always coexist.[18] The presence of hearing difficulties may limit their auditory feedback during voice practice. Because WBV requires relatively less degree of cognitive and coordinative skills, it can be more appealing to older adults. Moreover, newer generations of WBV devices available in the market come with a much smaller size and lower cost, making WBV a more affordable and accessible alternative intervention to vocal fatigue relief.

Research objectives and hypothesis

This study aimed to investigate whether the effect of WBV for relieving vocal fatigue 1) differs between young and older adults and 2) changes with vibration duration. It was hypothesized that WBV will be effective in relieving vocal fatigue in both young and older adults, but the extent of relaxation will be different between the groups as the vibration duration increases.

METHODS

The study was approved by the Faculty Research Ethics Committee, Faculty of Education at the University of Hong Kong (Reference number: EAU23030). Written consent was obtained from all participants prior to the experiment.

Participants

Eighteen older females who were 65 years of age or above (mean age = 65.56 years, SD = 4.44) and 18 young females who were aged between 20 and 25 years (mean age = 22.39, SD = 1.65) with healthy voices were recruited via the university's bulk email and word of mouth. All participants met the following inclusion criteria: 1) were vocally healthy on the experiment day; 2) had not received any formal vocal training before; 3) absence of other laryngeal and/or neurological pathologies as per self-report and auditory-perceptual judgment by the researcher; 4) had not received any medical diagnosis, medication, surgery, or voice therapy that might affect voice production within three months prior to participation; 5) did not have smoking and drinking habits; and 6) had adequate cognitive ability to complete experimental tasks. Participants with medical conditions unfit for WBV, such as those with cardiac pacemaker, epilepsy, vertigo, and recent bone fractures,[42] were excluded. To ensure eligibility, case history taking on general health and voice use history as well as cognitive screening with Mini-Cog[43] were done on the day of experiment.

Procedures

The entire procedure took place in a soundproof room in the Voice Research Laboratory at the University of Hong Kong. Vocal fatigue was induced with a karaoke singing task, adopted from Yiu et al.[22] Participants were required to sing karaoke on their own continuously for at least 95 minutes against a background noise of 60 dB. Participants were not allowed to drink any water throughout the singing task. After continuously singing for 95 minutes, participants were asked to rate their perceived level of vocal fatigue using a 5-point scale (1 = not fatigue at all, 5 = extremely fatigue). A minimum of four points was needed to be considered sufficiently fatigued, otherwise singing would continue until the criterion was met. Once the participant reported perceived vocal fatigue level of four points or above, participants were required to sing one additional song so that vocal fatigue was guaranteed.

After vocal fatigue induction, participants received WBV treatment for a total of 15 minutes. The WBV machine used was TT2590 (Turbosonic Ovation, Hood River, United States). The vibrational amplitude was fixed as 1 mm, while the vibrational frequency was tailored to each participant. According to the WBV treatment protocol developed by Yiu et al,[22] the optimal vibrational frequency should create the strongest feeling of vibration at the larynx and at a level of acceptable comfort.

Outcome measures

Three outcome measures were used to show any changes and improvements in phonatory functions: the highest fundamental frequency produced (objective measure), self-perceived vocal effort, and self-perceived vocal fatigue level (subjective measures). These measures were chosen based on previous similar studies on WBV and vocal fatigue[22,23] as well as their sensitivity in capturing the experience of vocal fatigue. A reduction in the highest fundamental frequency following acute vocal loading has shown in previous studies.[44–46] In order to keep track of participant's phonatory functions, measurements were done at six timepoints: prevocal loading, postvocal loading, post 5-minute WBV (or WBV5), post 10-minute WBV (or WBV10), post 15-minute WBV (or WBV15), and 15 minute postWBV completion.

The highest fundamental frequency

The highest fundamental frequency (highest F0) was elicited in a glissando manner during which participants produced /a/ from conversational pitch to their highest possible pitch.[22] Three trials were conducted with only the best performance counted.[47] Prior to prevocal loading measurement, participants were given at least three practice trials and were coached with demonstration from the researcher and visual feedback using the Singscope App on a tablet. During actual measurement, verbal coaching and gestural cue of hand-sweeping were provided.[48] Participants' production was recorded by a stand-alone dynamic microphone (SM48, Shure, Dubai, United Arab Emirates), with 8-cm mouth-microphone distance, through an interface (M-track, M-audio, Rhode Island, United States) onto an audio software (Audacity, version: 3.4.2).

Self-perceived vocal effort and self-perceived vocal fatigue level

Two self-rating scales were used respectively. The first scale was the 10-point OMNI vocal effort scale[49] that measured the amount of self-perceived vocal effort. The second scale consisted of six selected items (5-point scale) from the Hong Kong-Chinese Version of the Vocal Fatigue Index (VFI)[50]. The second scale examined the frequency of vocal fatigue symptoms and measured participant's self-perceived vocal fatigue level. In both scales, a higher score implied higher level of vocal fatigue.

DATA ANALYSIS

Statistical analyses were performed using IBM SPSS Statistics software (version 28.0.1.0). This study consisted of one between-subject factor: age with two levels (young and older adults) and one within-subject factor: vibration duration with six levels (prevocal loading, postvocal loading, WBV5, WBV10, WBV15, and 15 minutes after WBV treatment completed). The effect of vocal loading task was indicated by the difference in prevocal and postvocal loading measurement, while the relaxation effect of WBV was tracked with the remaining four measurement points. Changes across age and vibration duration were analyzed using two-way mixed analysis of variance (ANOVA). Pairwise comparison was adjusted with Bonferroni adjustment. Normality tests were conducted and the results indicated that the sample fulfils the requirement of using parametric tests. The statistically significant level was defined as $P < 0.05$.

RESULTS

Demographics of participants

Table 1 presents participant's characteristics, including age, height, weight, body mass index (BMI), and vibrational frequency (Hz). Independent sample t tests revealed significant difference in height ($t = 2.50$, $P < 0.01$), BMI ($t = -2.56$, $P < 0.05$), and vibrational frequency ($t = 3.78$, $P < 0.01$) but not in weight ($t = -0.76$, $P = 0.46$). Correlations between vibrational frequency and the other variables were explored using Pearson's r. Vibrational frequency was significantly positively correlated to height ($r = 0.39$, $P < 0.05$) and negatively and moderately correlated with age ($r = -0.58$, $P < 0.01$), but not significantly correlated to weight ($r = 0.04$, $P = 0.80$) or BMI ($r = -0.23$, $P = 0.18$).

Induction of vocal fatigue using Karaoke singing

On average, older adults and young adults sang karaoke for 105.17 minutes (SD = 11.29; range = 98-145 minutes) and 103.39 minutes (SD = 4.97; range = 98-115 minutes), respectively. There was no significant difference in mean singing duration between groups ($t = -.61$, $P = 0.55$).

The means and standard deviations of the three outcome measures for each group are shown in Table 2. The measurement taken at presinging served as the baseline vocal performance. The difference between presinging and postsinging indicated the effect of vocal loading. Prior to singing, a significant group difference was noted for the highest F0 ($t = 4.30$, $P < 0.05$), but not for self-perceived vocal effort ($t = 1.06$, $P = 0.30$) or perceived vocal fatigue level ($t = 1.62$, $P = 0.11$).

Upon completion of vocal loading task, the highest F0 produced had not changed significantly in both older ($P = 1.0$) and younger adults ($P = 1.0$). Nevertheless, both groups showed significant increase in self-perceived vocal effort and vocal fatigue level (both $P < 0.05$).

Changes in vocal functions, self-perceived vocal effort, and vocal fatigue across time

Figure 1 shows the mean change of the highest fundamental frequency, self-preceived vocal effort, and self-preceived vocal fatigue levels across the six timepoints. Two-way repeated measures ANOVA was conducted with post

TABLE 1.
Characteristics (Means and Standard Deviations) of Older Adults and Young Adults

Group		Age (in years)	Height (in cm)	Weight (in kg)	BMI (in kg/m^2)	Vibrational Frequency (in Hz)
Older adults	Mean (SD)	65.56 (4.44)	153.96 (4.79)	56.85 (7.67)	24.01 (3.38)	9.14 (2.06)
Young adults	Mean (SD)	22.39 (1.65)	159.02 (7.12)	55.00 (7.00)	21.70 (1.78)	11.08 (0.71)

Abbreviations: BMI, body mass index.

TABLE 2.
Mean (and Standard Deviation) of the Highest Fundamental Frequency Produced, Self-Perceived Vocal Effort, and Self-Perceived Vocal Fatigue Level Across Timepoints

Timepoints	Highest Fundamental Frequency (in Hz)		Self-Perceived Vocal Effort		Self-Perceived Vocal Fatigue Level	
	Older	Young	Older	Young	Older	Young
Prevocal loading	681.07 (201.2)	1130.26 (394.8)	1.22 (1.48)	1.78 (1.67)	4.17 (3.60)	6.39 (4.57)
Postvocal loading	734.42 (189.3)	1125.45 (289.5)	5.94 (1.43)	6.78 (2.04)	12.44 (4.31)	14.94 (4.70)
WBV5	673.79 (199.6)	1196.74 (310.3)	4.56 (1.58)	5.33 (1.85)	9.44 (4.19)	10.39 (5.12)
WBV10	687.18 (178.8)	1223.40 (335.0)	4.06 (1.92)	3.83 (1.65)	8.28 (5.01)	7.44 (4.84)
WBV15	757.47 (225.2)	1156.49 (290.7)	3.61 (2.12)	2.83 (2.18)	8.00 (4.69)	5.72 (5.02)
15 minutes postWBV completion	737.73 (230.0)	1202.44 (330.0)	3.33 (2.25)	2.44 (2.26)	6.89 (4.48)	5.11 (5.50)

Notes:
The maximum possible scores for self-perceived vocal effort and vocal fatigue level are 10 and 24, respectively. The higher the scores, the greater level of self-perceived vocal effort and vocal fatigue.
WBV5 = received WBV for 5 minutes; WBV10 = received WBV for 10 minutes; WBV15 = received WBV for 15 minutes.

hoc test. For the highest F0, repeated ANOVA showed a significant main effect of age ($F = 31.77$, $P < 0.001$, $\eta_p^2 = 0.48$) in which the values for older adults were consistently lower than young adults. However, no significant effect of vibration duration ($P = 0.79$) or interaction effect (age x vibration duration) ($P = 0.14$) was found.

Both self-perceived vocal effort and vocal fatigue levels showed similar trend and hence were reported jointly. There was a significant main effect of vibration duration for self-perceived vocal effort ($F = 73.01$, $P < 0.001$, $\eta_p^2 = 0.68$) and vocal fatigue levels ($F = 48.36$, $P < 0.001$, $\eta_p^2 = 0.80$). There was also an interaction effect (age x vibration duration) for vocal effort ($F = 3.87$, $P = 0.017$, $\eta_p^2 = 0.10$) and vocal fatigue levels ($F = 5.25$, $P = 0.004$, $\eta_p^2 = 0.29$).

Table 3 lists the results of post hoc test on self-perceived vocal effort and vocal fatigue level. The mean difference between prevocal loading and other subsequent timepoints indicates the proximity of a particular time point to the baseline vocal function. For self-perceived vocal effort, older adults showed significant difference between prevocal loading and postvocal loading, prevocal loading and WBV5, prevocal loading and WBV10, as well as prevocal loading and WBV15. On the other hand, young adults only showed significant difference between prevocal loading and postvocal loading, prevocal loading and WBV5, as well as prevocal loading and WBV10. As for self-perceived vocal fatigue level, both older and young adults showed significant difference between prevocal loading and postvocal loading as well as prevocal loading and WBV5.

The mean differences between consecutive timepoints are also reported in Table 3. It was used for tracking the progress of vocal recovery throughout and after WBV. For both self-perceived vocal effort and vocal fatigue levels, older adults only showed significant difference between postvocal loading and WBV5, whereas young adults showed significant difference between postvocal loading

TABLE 3.
Mean Difference of Post Hoc Test for Self-Perceived Vocal Effort and Self-Perceived Vocal Fatigue Levels

Point (a)	Point (b)	Self-Perceived Vocal Effort		Self-Perceived Vocal Fatigue	
		Older	Young	Older	Young
Prevocal loading	Postvocal loading	−4.72**	−5.00**	−8.28**	−8.56**
Prevocal loading	WBV5	−3.33**	−3.56**	−5.28*	−4.00*
Prevocal loading	WBV10	−2.83*	−2.06**	−4.11	−1.06
Prevocal loading	WBV15	−2.39*	−1.06	−3.83	0.67
Prevocal loading	15 minutes postWBV completion	−2.11	−0.67	−2.72	1.28
Postvocal loading	WBV5	1.39**	1.44**	3.00*	4.56**
WBV5	WBV10	0.50	1.50**	1.17	2.94**
WBV10	WBV15	0.44	1.00*	0.28	1.72*
WBV15	15 minutes postWBV completion	0.28	0.38	1.11	0.61

Notes: The mean difference is obtained by point (a) subtracting point (b).
The maximum possible scores for self-perceived vocal effort and vocal fatigue are 10 and 24, respectively. The higher the scores, the greater level of self-perceived vocal effort and vocal fatigue.
WBV5 = received WBV for 5 minutes; WBV10 = received WBV for 10 minutes; WBV15 = received WBV for 15 minutes.
*$P < 0.05$; **$P < 0.001$.

and WBV5, WBV5 and WBV10, as well as WBV10 and WBV15.

DISCUSSION

WBV has been used as an alternative method to relieve vocal fatigue. The present study evaluated the effects of WBV for relieving vocal fatigue in young and older adults. Specifically, it explored how age and vibration duration may influence the relaxation effect of WBV. The hypothesis was that WBV would be effective in relieving vocal fatigue in both older adults and young adults, but the extent of relaxation would be different between the groups as vibration duration increased. The three outcome measures were the highest fundamental frequency, self-perceived vocal effort (measured by the OMNI vocal effort scale), and self-perceived vocal fatigue level (measured by six items from the VFI).

Use of prolonged Karaoke singing task to induce vocal fatigue

In this study, prolonged karaoke singing task was used to induce vocal fatigue. Prior to singing, both older and young adults showed similar self-perceived vocal effort and vocal fatigue level. However, the highest F0 was significantly lower in older adults as compared with young adults. Such an age difference in the highest F0 aligns with previous studies and could be explained by the age-related laryngeal changes such as increased stiffness of vocal ligaments and laryngeal muscles atrophy.[51,52] The singing task was effective in inducing vocal fatigue as a significant pre-post vocal loading increase was noted for self-perceived vocal effort in both older (mean difference = 4.72) and young adults (mean difference = 5.00). Similar increase was observed for vocal fatigue level in both the older (mean difference = 8.28) and the young (mean difference = 8.56). The results also indicated that older and young adults required similar singing duration to reach vocal fatigue, with a mean of 105.17 and 103.39 minutes, respectively. Such durations were comparable to previous studies that reported a singing duration of 107.6[23] and 108.88 minutes.[22] The current study provides further support to the prolonged karaoke singing protocol and finds it effective when applied to the older adult population.

The effects of WBV for relieving vocal fatigue in older and young adults

Upon receiving WBV for five minutes, both older and young adults showed a significant reduction in self-perceived vocal effort and vocal fatigue level. There was no significant main effect of age for these two outcome measures. Both groups showed comparable increase of vocal fatigue after vocal loading and decrease after five minutes of WBV. On the contrary, the highest F0 showed no significant changes between postvocal loading and WBV5 in either group. The absence of change in the highest F0 despite significant self-perceived improvement is different from the results in previous studies that showed an improvement in the highest F0 after WBV.[22,23] One possible reason leading to the disparity may be the difference in elicitation method across studies. In the studies by Yiu et al,[22,23] the highest F0 was recorded using phonetogram that provides real-time visual feedback. A note imitation procedure was also used after each pitch glide to ensure that the highest F0 was reached. Contrastively, the current study only used hand-sweeping gestural cue when participants performed pitch glide with no simultaneous auditory or visual feedback during measurement. The literature has indicated that visual feedback is useful in boosting participant's maximum vocal performance and also for enhancing motivation.[53,54] The absence of visual feedback may render the risk of not extending participants' vocal range to their widest level, hence reducing the capacity of highest F0 in capturing phonatory changes in the current study.

The effects of WBV with increasing vibration duration

Both older and young adults showed a significant main effect of vibration duration and an interaction effect (age x vibration duration) for self-perceived vocal effort and vocal fatigue level. In other words, as vibration duration increased, different vocal recovery pattern was observed between the age groups. From post hoc comparison, it was found that young adults continued to show significant reductions in self-perceived vocal effort and vocal fatigue level when vibration duration increased from 5 to 10 and then 15 minutes. However, older adults did not show this continuous recovery and it appeared that the relaxation effect for older adults started to slow down at 10 minutes of WBV. When comparing the intervention timepoints to the prevocal loading baseline for self-perceived vocal effort, it was noted that young adults returned to baseline after receiving 15 minutes of WBV. However, the older adults achieved so only until 15 minutes after the WBV finished, which was later than the young adults. To summarize, findings from both self-perceived vocal effort and self-perceived vocal fatigue level indicated a slower recovery in older adults as compared with young adults when duration of WBV increased.

The above findings agree with the hypothesis of the current study that older and young adults experienced vocal fatigue relief during WBV to a different extent. Specifically, older adults showed a slower rate of fatigue recovery with increased vibration duration. From a neuromuscular perspective, fatigue could be categorized into central and peripheral fatigue. Central fatigue refers to the decrease in voluntary neural firing to the muscle, while peripheral fatigue refers to the strength reduction of muscle fiber contraction.[55,56] These two processes can be used to explain the age effect on the recovery pattern observed in this study.

Concerning central fatigue, older adults experience functional decline in their neuromuscular system and remodeling of motor units.[57] Research in muscle physiology

has attempted to study the difference in fatigue recovery in young and older adults from the lens of central fatigue. For instance, Hunter et al[58] examined the maximum voluntary contraction of elbow flexor muscles and older adults had a significantly lower voluntary activation at the recovery stage. Dalton et al[59] studied the soleus muscle following high-intensity isometric exercise, with older adults showing a slower recovery of firing rate of motor units. Solianik et al[60] investigated the knee flexor muscles from maximal voluntary contraction. A significant correlation between central activation ratio and maximal voluntary contraction recovery was noted in older adults. In short, these findings from skeletal muscles point out that older adults have a higher level of central fatigue compared with young adults, which contributes to their slower fatigue recovery.

As for peripheral fatigue, it concerns the consumption of cellular energy and an accumulation of metabolic waste.[55,61] Specifically, the production of lactic acid, a metabolic by-product from anaerobic respiration, is closely linked to muscle fatigue because it impedes effective muscle contraction.[29,62,63] The capacity to resist and recover from peripheral fatigue in older adults can be impacted by various age-related changes in muscle composition, blood flow, and metabolism.[64] Previous research showed that older adults produce less lactic acid upon exercise.[65,66] A reason behind is that older adults utilize the mechanism of oxidative phosphorylation more than anaerobic glycolysis when generating ATP for cellular activity.[67,68] Though older adults have lower lactate accumulation compared with young adults, their removal rate of blood lactate is slower due to the reduced blood flow during and after exercise. Laryngeal blood flow has been found to be reduced significantly with age in rats, with the greatest reduction in the thyroarytenoid muscles.[69,70] Russell et al[71] examined the effects of aging on laryngeal blood flow in senescent rats. They found significant age-related vascular changes in microcirculation of the thyroarytenoid muscles. In short, with a slower blood circulatory system and vascular changes in microcirculation, it is predictable that older adults dispose lactate in a slower rate compared with young adults and hence their rate of fatigue recovery is also slower.

Limitations and future research directions

The following limitations should be acknowledged when interpreting the current results. First, the current study allows a comparison of vocal fatigue relief between age groups after WBV. As vocal recovery occurs naturally without intervention, future study should include a control group with vocal rest only. This can help validating the observed benefits from WBV, and confirm that the observed differences between young and older groups are not simply attributed to voice rest and time. Apart from a voice rest control group, additional alternative treatment controls such as localized laryngeal vibration and semi-occluded vocal tract exercises should also be included. This provides a more complete understanding and justification for using WBV as an intervention modality for vocal fatigue in the older population. Second, the current study adopts only the highest F0 produced, self-perceived vocal effort, and self-perceived vocal fatigue level as the outcome measures. The rationale for using a small number of outcome measures is to allow a quick and noninterfering measurement during the WBV intervention. Nevertheless, one should be cautious that the reliance on self-reported measures to track the recovery from the vocal loading task may pose potential of observer-expectancy effect (ie, participants' self-bias in reporting of fatigue level as a result of being participating in the intervention). Further studies should consider incorporating more objective acoustic and aerodynamic measures in tracking the recovery from the vocal loading task. Measures of cepstral peak prominence, low/high spectral ratio, and phonatory threshold pressure have shown to be sensitive to subtle laryngeal changes.[72] The inclusion of placebo intervention and also diverse objective measurements would strengthen the interpretation of self-reported measures. Finally, in this study, the inclusion criteria of a normal voice were based on participants' self-report and auditory-perceptual judgment by the researcher. Because presbylarynx can exist in older individuals without obvious voice complaints, visual examination of the larynx is warranted to screen out any presence of laryngeal abnormalities as this may have implications on how participants recover from the vocal loading task.

CONCLUSION

This study compares the effect of WBV for vocal fatigue relief in both young and older adults as well as tracks the recovery progress with increasing vibration duration. The results suggest a significant main effect of vibration duration and an interaction effect (age x vibration duration) for both self-perceived vocal effort and vocal fatigue level. Both older and young adults experienced significant reduction in self-perceived vocal effort and fatigue level at 5 minutes of WBV, but the effect starts to slow down for older adults at 10 minutes of WBV. The highest F0 produced showed a significant main effect of age only, but not main effect of vibration duration, which could possibly be due to elicitation procedure. The results provide support that WBV may be used in both older and younger adults for vocal fatigue relief. Future studies with control groups and additional outcome parameters are warranted to establish the effects and optimal intervention condition of WBV in older adults.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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